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METHODOLOGY PAPER · WORKING WITH CLAUDE CODE

# One Coordinate, Three Days.

*How an analyst and Claude Code turned a single GPS point into a repeatable satellite-imagery screening workflow, with the exact prompts, the conversations that mattered, and a playbook for doing it again.*

Subject: the Yemen earthwork project, 21–23 September 2026 (Tihama coast, Marib, Sept 2026 front)
Prepared for: colleagues and leadership who want to understand or start working this way · Date: 29 September 2026 · Method: review of 12 Claude Code session logs, per Appendix D
Project owner: GitHub `bhngyn` · Drafted by: Claude (Sonnet 5.5) from the logs, at the owner's request

## THE HEADLINE, IN A PARAGRAPH

Between the evening of 21 September and the night of 23 September 2026, one analyst and Claude Code went from **one coordinate on the Red Sea coast of Yemen** to a working method for finding, dating and reviewing military earthworks in public satellite imagery. The record is **12 logged sessions, 41 typed prompts (median 17 words) and 989 tool calls**, roughly 24 actions per prompt, with no subagents. The work produced two trench complexes on the Tihama coast and two more at Marib, each confirmed and dated on sub-meter imagery, a region-agnostic screening package (`trenchkit`, just under 1,000 lines), ten offline HTML review documents, twelve KML/KMZ review files for Google Earth, and re-applications to the September 2026 front and to Marib.

The practices that made it work were unglamorous: a **concrete anchor** in every request, the analyst's **hand-drawn labels as the yardstick**, "**ask me questions**" at design forks, **reviewable files** instead of chat summaries, and **memory notes** so a new session could resume without re-briefing. The mistakes were just as instructive, and they are cataloged in §7.

*This paper documents a method and its record. It does not independently validate the imagery findings; the confidence statements quoted here were Claude's assessments at the time, and attribution of who built what is analytic judgment at moderate to low confidence.*

| SESSIONS | TYPED PROMPTS | TOOL CALLS | ELAPSED | MEMORY NOTES | SUBAGENTS |
|---|---|---|---|---|---|
| 12 | 41 | 989 | 50 h | 10 | 0 |
| one is a 2-turn fork | median 17 words | ≈24 per prompt | first to last message | carried across sessions | parallel sessions instead |

### FOR LEADERSHIP: SIX TAKEAWAYS

1. **Pilot with an anchor, not a specification.** The project began with one coordinate and a data request; the scope grew as results arrived.
2. **Invest in ground truth.** The scarce input was an analyst's 185 km of hand-drawn labels, not computing power. It is what made the detector measurable.
3. **Require reviewable outputs.** Every deliverable opened in the reviewers' own tools and carried a list of what was not verified.
4. **Keep human sign-off on irreversible or outward-facing steps**: deleting, publishing, and attributing who built what.
5. **Count setup as part of the work.** Data access, environment notes and saved memory were unglamorous and are what let later sessions start fast.
6. **Decide the publication policy for sensitive locations before sharing** (§9). Here the owner chose to keep exact coordinates and attribution; the reasoning is in §9.


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## 1. The project in one page

The project applies open-source imagery to the Yemen conflict. The starting question was narrow: local reports said fighters were digging trenches through farmland on the Tihama plain near Zabid, and a coordinate on the coast marked where one of them began. Was that visible from space, and were there others?

Over three days that question grew into a pipeline. Free radar (Sentinel-1) and optical (Sentinel-2) satellite data were used to *screen* large areas for new straight-line features. Esri's Wayback archive of sub-meter imagery was then used to *confirm and date* the candidates by eye. The analyst hand-traced 185 km of trenches in Google Earth, and those labels became the test that told Claude which signals worked and which did not. The result was packaged so it could be pointed at a new region.

| WHAT WAS PRODUCED | IN PLAIN TERMS | CONFIDENCE STATED AT THE TIME |
|---|---|---|
| **Site B**, Al-Fazah | A 5.6 km ditch running from the beach inland, built in two campaigns (Mar–Dec 2022; Jun 2023–Nov 2024). | Military obstacle: almost certain. Builder: very likely Houthi, moderate confidence. |
| **Site A**, Al-Jarrahi–Hays road | A roughly 9 km trench complex beside the highway, absent in Oct 2021 and complete by Apr 2025. | Military position: almost certain. Builder as above. |
| **trenchkit** | A region-agnostic screening pipeline: define a region, fetch imagery, detect, rank, then review on an offline page with sub-meter frames. | Baseline recorded: top-100 candidates contain 19 % of the analyst's labeled kilometers. |
| **Sept 2026 front** | About 11 km of new or re-graded vehicle tracks on Jabal Kahbub, appearing between the 11 and 16 Sep satellite passes. No large new earthwork elsewhere. | Real ground change: high. Who built it: low. |
| **Original area rerun** | No new digging at sites A/B or any labeled trench after the 10–11 Sep offensive; one freshened 1.3 km track on the old front. | Moderate for the track; "no change" holds for B-sized works only. |
| **Marib** | Two complexes: an 8.4 km ditch network at Raghwan and an ~11 km straight earth bank north of the city, both dated to windows of 1–3 years. | Confirmed at 0.3–0.6 m; builder not established. |
| **Review tools** | Ten offline HTML documents and 12 KML/KMZ files so the analyst could check everything in familiar tools. | Not all opened in Google Earth (§9). |

> **WHAT THE SUMMARY HIDES**
>
> The method was not designed in advance. It **accreted**: each result made the next question obvious. The day-one prompts were requests for OSM canals and a Google Earth file; by the end of the second day the analyst was asking for a detection package that could be applied to "new regions in Yemen". Nothing in the first prompt says "build a screening pipeline." That is the main point of this paper: a small anchor, a fast loop, and an assistant that keeps state.

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## 2. Vocabulary in sixty seconds

Readers come from different backgrounds, so the two vocabularies that recur are defined once here. Both are also used in the quoted conversations in §6.

## CLAUDE CODE TERMS

| | |
|---|---|
| **Session** | One continuous conversation with Claude in a project folder. Each has an ID; this project has 12. |
| **Tool call** | One action Claude takes: run a command (Bash), read a file (Read), search the web, send you a file. |
| **Memory note** | A small text file Claude writes to remember facts across sessions (environment quirks, project status, your preferences). |
| **Skill** | A reusable instruction pack, e.g. a house style for reports. Invoked by name. |
| **Background job / Monitor** | Long downloads or computations run while Claude keeps working; Monitor wakes it when something happens. |
| **Ask-questions form** | A multiple-choice prompt Claude can show you to settle a design decision. |
| **Fork** | A copy of a conversation continued separately. |

## IMAGERY TERMS

| | |
|---|---|
| **Sentinel-2 (S2)** | Free optical satellite, 10 m pixels, revisit every few days. Cloud-limited. |
| **Sentinel-1 (S1)** | Free radar satellite. Sees through cloud; bright where ground is rough or angled, such as fresh berms. |
| **Esri Wayback** | An archive of Esri's world imagery releases, often 0.3–0.6 m. Each image has a *capture date* that differs from the *release date*. |
| **Earth Engine (EE)** | Google's platform for processing satellite archives in the cloud. |
| **KML / KMZ** | File formats Google Earth opens: lines, pins, folders, notes. |
| **Recall at 0.5 % false alarms** | Of the true trench length, what share does a map find when only 0.5 % of non-trench ground is flagged. |
| **dB change** | Radar brightness difference between two dates, in decibels. |

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## 3. Timeline: five phases

The record covers 50 hours of wall-clock time, from the first prompt on 21 September at 20:44 UTC to the last on 23 September at 22:31 UTC.[1] Sessions sometimes ran side by side (two were open together for over an hour on the morning of 22 September, UTC), which is how parallel work happened without subagents. The first eight sessions and the start of the ninth ran on **Claude Fable 5.1**; the analyst switched to **Claude Opus 5.5** with `/model` at 22:41 UTC on 22 September, and everything after ran on it.

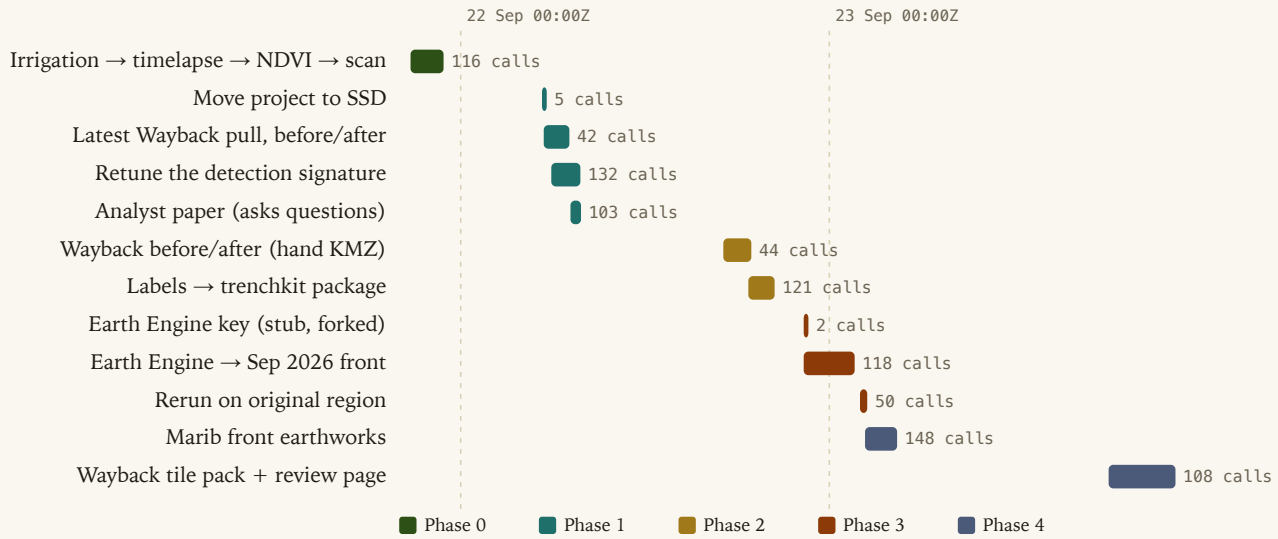



*Figure 1. Every logged session as a bar from first to last message (UTC), with tool-call counts. The Marib session is a single prompt that ran for about two hours, much of it waiting on Sentinel-2 composites that took about 30 minutes per year to download.*

| PHASE | WINDOW (UTC) | SESSIONS | TYPED PROMPTS | TOOL CALLS |
|---|---|---|---|---|
| **Phase 0: one coordinate to a candidate map** | `21 Sep 20:44Z → 21 Sep 22:53Z` | 1 | 10 | 116 |
| **Phase 1: retune, review, first analyst paper** | `22 Sep 05:17Z → 22 Sep 07:49Z` | 4 | 12 | 282 |

| PHASE | WINDOW (UTC) | SESSIONS | TYPED PROMPTS | TOOL CALLS |
|---|---|---|---|---|
| **Phase 2: the analyst's hand labels** | `22 Sep 17:05Z → 22 Sep 20:25Z` | 2 | 2 | 165 |
| **Phase 3: Earth Engine, the September front** | `22 Sep 22:19Z → 23 Sep 02:27Z` | 3 | 8 | 170 |
| **Phase 4: Marib and the review tool** | `23 Sep 02:18Z → 23 Sep 22:31Z` | 2 | 9 | 256 |

### Phase 0: from one coordinate to a candidate map (21 Sep, 20:44–22:53Z)

Ten prompts in a single session. The analyst asked for OSM irrigation layers within 50 km of a point, a KML of them, the World Bank's Tihama datasets, local reporting on trenches, a Sentinel-2 timelapse from 2021, a re-centered timelapse, an NDVI analysis, whether the trench itself was detectable, a scan of the whole 50 km circle, and finally: "there are major trenches and earthworks here" at a second coordinate. That last message changed the detector from optical-first to radar-first (Excerpt 3).

### Phase 1: retune, review, first analyst paper (22 Sep, 05:17–07:49Z)

Four sessions, two of them overlapping for over an hour. Claude fetched the newest Wayback imagery for every candidate, was asked to make the traces easier to see, then pulled older releases for before/after pairs. In another session, the analyst said Site B was "the clearest example of known constructed trenches", and Claude rebuilt the signature around it, publishing a report as a private claude.ai Artifact. The analyst then asked for offline HTML instead, and the next report, a military-analyst paper, was preceded by two rounds of structured questions.

### Phase 2: the analyst's hand labels (22 Sep, 17:05–20:25Z)

Two prompts, each carrying a KMZ the analyst had drawn by hand. The first requested Wayback before/after imagery along the labeled lines; the second asked Claude to learn from the labels, improve detection, package the process and "ask me questions to help design the process". This phase produced the `trenchkit` package and the evaluation numbers in Figure 4.

### Phase 3: Earth Engine and the September front (22 Sep 22:19Z – 23 Sep 02:27Z)

The analyst registered for Google Earth Engine, Claude got it working and pulled 868 radar chips (1.9 GB) with no failures. Then: "look at the new areas of fighting and look for signs of new construction." A rerun on the original area followed, plus a KML for review.

### Phase 4: Marib and the review tool (23 Sep, 02:18–22:31Z)

One prompt ("are there similar structures in the marib frontline?") ran a session of about two hours that ended in two dated finds. The final eight prompts, mostly short corrections, turned a folder of imagery into a 420 MB self-contained review page with in-browser downloads.

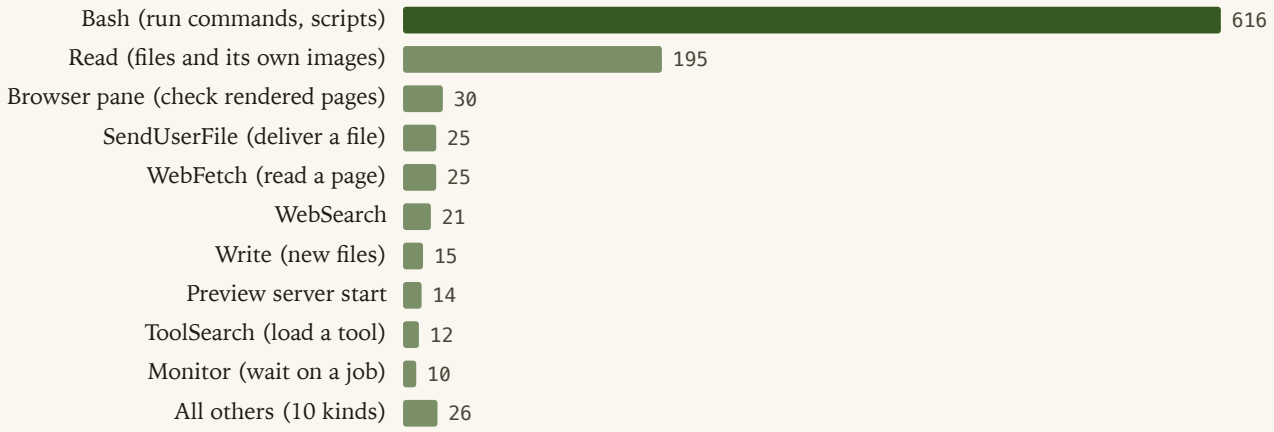



*Figure 2. What Claude actually did: 989 tool calls across all sessions. Running commands and reading files, including its own rendered images, account for four of every five calls. Only two structured question forms and one web-published page were used.*

## 4. The working loop

Beneath the variety, almost every exchange followed the same six-step loop. The analyst did two of the steps (framing and reviewing); Claude did the other four. Most laps began with a prompt of about 17 words.

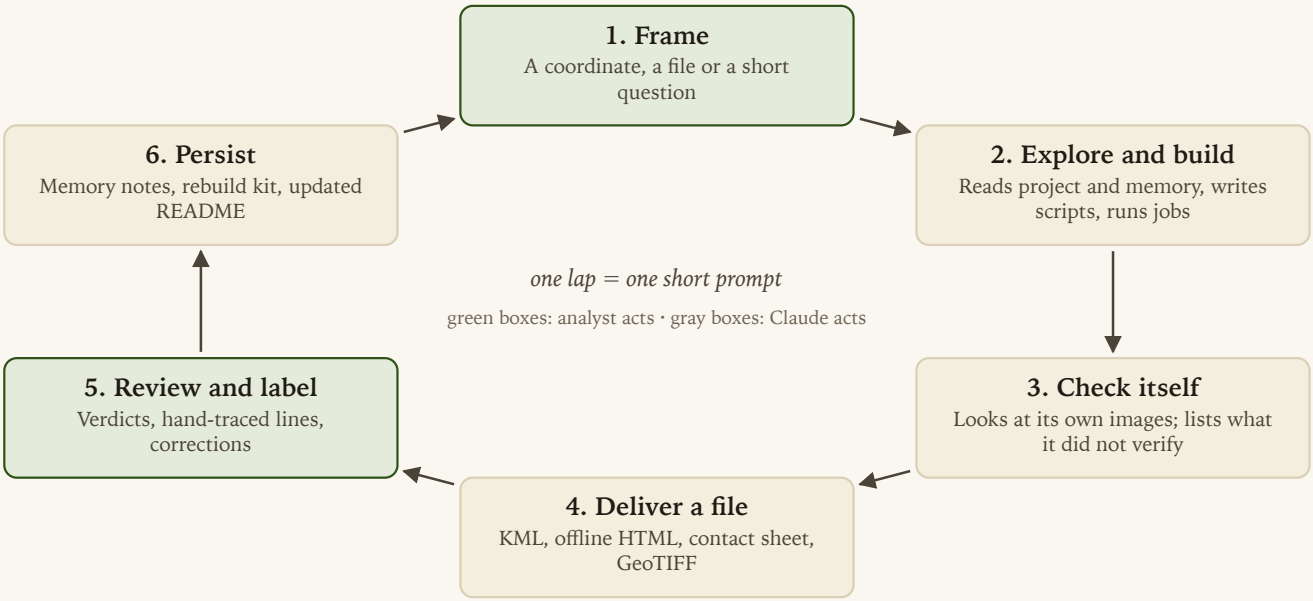



*Figure 3. The loop as observed. Step 3, checking itself, is where most of the quality came from: Claude opened its own contact sheets, found problems (a color-bar overlap, black mosaics, a mislabeled feature) and said so.*

| THE ANALYST SUPPLIED | CLAUDE SUPPLIED | NEITHER ALONE COULD DO |
|---|---|---|
| Anchors, ground truth, verdicts, taste | **Data plumbing, statistics, review tooling, writing** | Tune a detector against real labels overnight |

Two structural features are worth naming. First, the analyst almost never described *how*. Prompts said what they wanted to see ("show me on sentinel 2 what you find") and in what form ("create a kml for me to review"). Second, the review artifact was chosen to fit the analyst's own tools: KML for Google Earth, offline HTML with verdict boxes that export to JSON, GeoTIFFs for QGIS. The deliverable was the interface.

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## 5. Thirteen lessons and best practices

Each lesson is anchored in something that happened, with the session or prompt number so it can be checked against Appendix B.

### 1 Start with an anchor and let requests stack

The first prompt was a query for canals within 50 km of a coordinate. Nothing needed to be specified about the eventual goal. Ten short prompts later the project had a timelapse, an NDVI analysis, a dated

earthwork and a candidate map (prompts 1–10). The median prompt across the whole project was 17 words; the longest was 84.

**Do this:** begin with a location, a file or a single question. Each answer will suggest the next one. Write the long brief only when you have evidence for it (prompt 26).

## 2 Make Claude look before it builds

In nearly every session the first several actions were reads: memory notes, the project tree, existing scripts. When the analyst supplied a new KMZ, Claude wrote "There's an existing Wayback toolkit in the project. Let me read it so I reuse it rather than rewrite"; the same kit was later reused for the gap fill and the Marib front. For "rerun the latest iteration of the analysis on the original region I requested", Claude searched the earlier sessions' prompts to work out what those phrases meant, and then said how it had read them (Excerpt 10).

**Do this:** keep vague, reference-heavy prompts ("the original region", "the latest round") but make sure the project has notes Claude can resolve them against.

## 3 Use "ask me questions" at forks, then actually answer

Two prompts ended with that instruction (21 and 26). The first produced two structured forms with seven questions; the second produced a plain-text list of eight. The answers were short and sometimes were *actions*: asked about data access, the analyst registered for Earth Engine and asked how to find the key, which answered the question by doing (Excerpt 5). One form answer bounded the work: "write from what is on disk now, defer the other options to be done later after I review what's on disk."

**Do this:** ask for questions when a choice changes what gets built. Answer in chat, or reply with the next action. Note which questions stayed open; here the design questions sat in a memory note as "answers pending".

## 4 Ground truth is the currency; make Claude check your labels too

The analyst's early guidance was a single example ("B is the clearest example of known constructed trenches I am interested in"). Claude then discovered the hand trace was 50–100 m off and built a new centerline by snapping it to 0.3 m imagery: "Earlier numbers against the old trace understated the gains." The 185 km KMZ went further: it turned a set of hunches into an evaluation with numbers, and showed that optical red-band change found 1 % of the labeled kilometers while radar change found 34 % (Figure 4).

**Do this:** hand-label a small, representative example set early, in a format Claude can rasterize (KML/KMZ worked). Ask Claude to validate the labels against the best imagery before trusting scores.

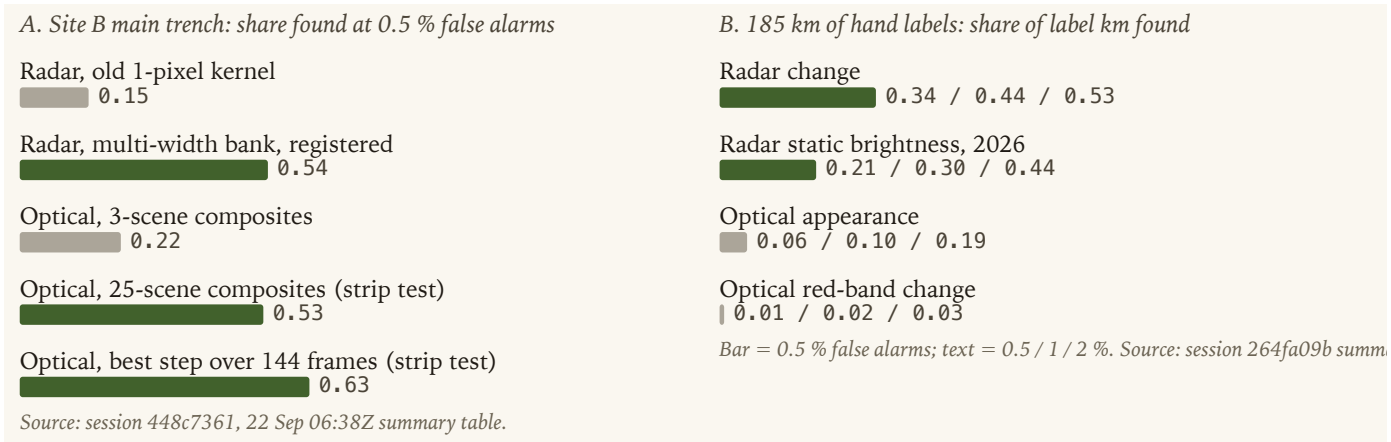



*Figure 4. Evidence the labels mattered. Panel A: retuning against Site B raised radar detection of its main trench about 3.5 times at the same false-alarm rate. Panel B: once the analyst's 185 km of labels arrived, the detector could be judged, and optical change turned out to be nearly useless for this feature type. Both panels are Claude's own reported figures from the sessions cited.*

## 5 Let measurements overrule the plan

Three times Claude abandoned its own approach because measurements against known sites or labels said so: an optical-first scan caught only a 1.3 km fragment of Site A, so radar became primary; fusing four maps before extraction lost weak lines, so extraction moved to a per-map union; and a learned ranker (13 clean positives, 24 after widening the definition) did not beat the hand-built score out of sample, so Claude stopped the running chain and made the ranker fall back automatically (Excerpt 6).

**Do this:** ask for an evaluation harness before optimization, and ask for "the baseline any change must beat" to be stored in a note.

## 6 Demand a statement of what was *not* verified

Claude's summaries repeatedly carried sentences such as "I haven't opened any of the layers in a GIS", "I haven't verified the Al-Mashjah rim feature at full resolution, so I'm removing that label from the map rather than claim it", and "couldn't open it in the browser here, so I haven't looked at the rendered page." It also retracted claims: it had named World Bank GIS datasets from memory, then searched and reported that none existed (Excerpt 2), and it corrected a feature it had mislabeled as a wadi channel (Excerpt 1).

**Do this:** treat unstated verification as unperformed. Ask "what did you not check?" at the end of a big deliverable. Prefer results that include a rejected-examples section.

## 7 Deliver the file that fits your tools

Claude made 25 file deliveries (SendUserFile calls) during the project. The pattern was consistent: KML for Google Earth (the analyst asked for one three times), contact sheets with the traces drawn thick enough to see, offline HTML with a verdict dropdown and notes per candidate and an "Export verdicts as

JSON" button, and GeoTIFFs for QGIS. One report went out as a claude.ai Artifact and the analyst redirected: "i want an offline html not an artifact" (recorded in a feedback note the same day).

**Do this:** say what the reviewer will open the result in. Ask for a rebuild script beside every report so it can be regenerated when data changes.

## 8 Put stable facts in memory, not in the next prompt

Ten memory notes were written: environment limits (which Python has the geo stack, 8 GB of RAM, slow downloads), project state (what sites A and B are, the front-line geography used for attribution), toolkit descriptions, the deliverable-format preference, and one note per major result. All seven substantive sessions that started after the first note was written read the notes within their first five actions (six of them as the very first action), and the radar pull for sites A and B started from the site coordinates found there.

**Do this:** after each major result, ask Claude to save a note with the facts a later session would need *and could not recover from the code*. Ask it to record deferred tasks explicitly.

## 9 Turn every repeatable step into a kit, and reuse house styles

Most reports have a builder script, a body template and a figure-prep script beside them. The Wayback fetch script gained `--release` and `--ids` arguments in its first session and was reused in later ones. The stack-download script was made idempotent so that rerunning it extends rather than repeats. Two skills supplied house style: `analysis-paper` (three sessions) and `tufte-graphics` (once, for the NDVI charts).

**Do this:** ask for "a script I can rerun" and for parameters at the top. Capture the report style once as a skill.

## 10 Match the data source to the job, and measure before you commit

Sentinel-1 downloads from one public source ran at 0.1–0.4 MB/s; Claude's own estimate was about a day per orbit for a 100 km region. Earth Engine delivered a 545 × 668-pixel chip in 1.3 seconds, returned all 868 chips without failures, and later fetched Marib composites at roughly three minutes per orbit. Before the big pulls Claude timed a test read (first a public-catalog window, then one Earth Engine chip).

**Do this:** ask for a throughput test on a small window before any multi-gigabyte pull. Record measured speeds in a note.

## 11 Run long jobs in the background, and control the noise

The harness blocks foreground `sleep` chains (four sessions hit that), so Claude used background jobs and the Monitor tool. One monitor was set to report every 500 tile hashes and produced 88 notifications

in a single session, most answered with a bare "Waiting." A later monitor that reported only start, finish or error was far quieter.

> **Do this:** tell Claude to monitor for *completion or failure*, not for progress. Prepare the next stage while a job runs.

### 12 Short corrections work when the cause gets found

"i only see a black screen" (six words) led Claude to work out that the analyst had opened the unfilled template rather than the built page, rename the template to a non-HTML extension and add a warning message. "I want the download to actually download" led to the reason (Chrome ignores the `download` attribute for local files it can display) and a fix (build a GeoTIFF inside the page) (Excerpts 8 and 9).

> **Do this:** report the symptom, not a solution. Expect Claude to explain a cause; if it only patches, ask "why did that happen?"

### 13 Keep irreversible and outward-facing decisions with the human

When moving the 8.4 GB project to an external SSD, Claude verified the copy by checksum and file count and then stopped: "The original ... is still in place. Deleting it is irreversible, so I did not do it on my own." Reports carried confidence tiers, not verdicts. Attribution was labeled geography-based and moderate or low. Claude also declined to treat an unverified social-media claim about Al-Jawf berms as evidence.

> **Do this:** state which actions need your OK (delete, publish, send, spend). Ask for confidence levels and their basis in every assessment.

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## 6. Key conversations, verbatim

Ten exchanges chosen because they show a decision, a correction or a technique. User text is exactly as typed (spelling included). Claude's text is trimmed for length with "…" and has decorative kaomoji removed.[2] Numbers refer to the prompt ledger in Appendix B.

EXCERPT 1 **Self-correction on a mislabeled feature** `session d86eefd2 · 21 Sep 21:32Z · prompt 8`

ANALYST

```
is there a way to detect the building of the military trench with sentinel
imagery? it begins at the precise point i've indicated
```

CLAUDE

Yes, there is, and we may already have caught it. In the earlier frames there's a thin line running east from your crosshair that isn't there in 2021; I mislabelled it a "wadi channel" in the NDVI map. Let me test that properly with the 10 m data already on disk.

*[…dated the feature from clear scenes: first fresh soil 4 Feb 2022, L-shaped cut 19 Feb 2022, extensions through 2023…]*

I need to correct two things I said before. *Not a wadi channel:* in the NDVI map I called this line a wadi channel reaching the sea. That was wrong. It is this earthwork. *Vegetation loss matches the earthwork:* the roughly 100 ha vegetation-loss cluster I dated to 2022–23 lines up with its construction in space and time.

**Why it matters.** The analyst asked a yes/no feasibility question; Claude answered it with evidence from files it had already downloaded and openly revised an earlier conclusion. Later reports kept the same habit of listing corrections.

EXCERPT 2 **Retracting a claim made from memory** `session d86eefd2 · 21 Sep 20:45Z · prompt 3`

ANALYST

```
can you pull the world bank tihama irrigation datasets for me too
```

CLAUDE

*[first line of its reply: "Let me see what actually exists publicly (I named those from memory, so I need to verify)". Then four searches and a page fetch.]*

I can't pull them, because as far as I can tell no public World Bank GIS dataset of the Tihama canal network exists. I suggested it last turn from memory, and that was overstated, sorry. The searches turned up reports and coarse rasters, but no downloadable vector canal map.

*[…followed by a table of what does exist, whether each is usable in GIS, and three options ranked by effort; it asked before downloading anything.]*

**Why it matters.** The correction cost one turn. Left uncorrected, the analyst would have been sent after a dataset that does not exist. The habit of verifying its own earlier statements recurs throughout.

EXCERPT 3 **Teaching the detector by example** `session d86eefd2 · 21 Sep 22:16Z · prompt 10`

ANALYST

`there are major trenches and earthworks here – can you adjust your detection process to proritize things with a similar signature? 14.056300910076873, 43.431572797625435`

CLAUDE

*[…37 minutes of work: rendered a 6 km view in both sensors, stacked 20 radar scenes per period, wrote a radar-first detector, calibrated on the two known sites…]*

At your location the trenches are nearly invisible in 10 m optical imagery, but Sentinel-1 radar shows them clearly. They appear as thin continuous lines where the radar return rose by about 2.5 dB between 2021 and 2026, because fresh berms and cut faces reflect radar back strongly. […] So I changed the method: *Radar is now the primary sensor.* […] Both known sites check out. The detector recovers your site as a single 9.6 km line, which ranked first of all 364.

**Why it matters.** One coordinate and one sentence were enough to change the method. The recall check on the two known sites is what made the change trustworthy.

EXCERPT 4 **An "ask me questions" round, and the analyst's reply** `session c66378c1 · 22 Sep 07:08–07:16Z · prompts 21, 22–23 (form answers)`

ANALYST

`I want you to make an analysis of the sentinel 1, 2 and wayback imagery you've reviewed so far from the perspective of a military geospatial analyst. ask me questions to shape your analysis`

CLAUDE

*[read the memory notes and the whole project first, then showed two forms. Questions included: "What kind of analytical product should this be?", "How far should I go on attribution and context beyond what the imagery shows?", "Do you hold ground truth on who built or holds sites A and B?", "Should I spend session time on extra collection before writing?"]*

ANALYST (FORM ANSWERS)

"How far should I go on attribution…"="Attribute and assess intent" · "Do you hold ground truth on who built or holds sites A and B? […]"="No, assess from open sources" · "Should I spend session time on extra collection before writing?"="write from what is on disk now, defer the other options to be done later after I review what's on disk"

**Why it matters.** The questions were grounded in project files Claude had just read, so the options were real choices. The free-text answer to the last question bounded the work; the paper was written from data on disk and the deferred collection was recorded in the paper's §7 and in memory.

EXCERPT 5 **The design brief, eight questions, and an answer by action** session 264fa09b · 22 Sep 18:43Z–20:25Z · prompt 26, then 27–35 in another session

ANALYST

@"/Users/brian/Downloads/Yemen identified trenches.kmz"
I've hand labeled and identified more trench systems in this region. Can you review this KML file and improve the signatures you use to detect these earthworks? You may need multiple signature types. Use your best judgment – I want to apply lessons learned here to new regions in Yemen. I want you to improve the signatures, improve detection, and then package the process so I can easily implement it in other regions. Ask me questions to help design the process

CLAUDE

*[100 minutes later, after building and testing a package on a fresh 3 km area:]*

Your 185 km of labels showed that the September 21 detector saw about a third of the trench belt, and that the missing two thirds are narrow fighting trenches that 10 m optical data cannot separate from clutter. I rebuilt the detector around what the labels support, packaged it as a region-agnostic pipeline, and verified the package end to end on a fresh 3 km test area. […] *Questions for designing the process:* 1. Target: only works dug between two dates, or everything present now? […] 3. Data access: the radar stack is the bottleneck. Do you have, or would you get, a Google Earth Engine account? With it the fetch stages become minutes […] 7. Packaging: keep the CLI and README, or also wrap it as a Claude Code skill so future sessions run it identically? […]

ANALYST (A DIFFERENT SESSION, TWO HOURS LATER; FOUR SEPARATE PROMPTS WITHIN SIX MINUTES)

```
how can i get to my google earth engine api key
I just registered
[GCP_PROJECT_ID]
pull Sentinel-1 stacks over sites A and B
```

**Why it matters.** The eight questions were never answered in text. Question 3 was answered by registering and asking for the key; Claude corrected the premise ("Earth Engine doesn't hand out an 'API key' at all") and finished the setup. The other seven are not answered anywhere in the logs, and Claude noted the questions in memory as "answers pending".

EXCERPT 6 **Claude stops its own pipeline when the evaluation disagrees** `session 264fa09b · 22 Sep 20:13Z`

CLAUDE (MID-RUN STATUS MESSAGE)

The learned ranker is not yet useful: only 13 components qualify as clean positives, and leave-one-complex-out it does no better than the hand score. I'm stopping the chain before the Wayback pulls use the worse ranking, making the ranker fall back to the hand score when validation shows no gain, and restarting from that point.

**Why it matters.** No one asked for this. The pipeline was running unattended, the next stage would have pulled Wayback imagery for the top 20 of the worse ranking, and Claude caught it. It then made the fallback automatic, so the package reports its own limits in future regions.

EXCERPT 7 **An assessment with three confidence tiers** `session 3981a117 · 23 Sep 01:38Z · prompt 38`

ANALYST

`summarize for me your assessment based on the sentinel 1 data analysis`

CLAUDE

The Sentinel-1 radar shows no new construction on either front that stands out from normal seasonal change. The radar did not find the Kahbub tracks either. Those came only from Sentinel-2. […] *Confidence.* High: no large new earthwork, such as a long berm or trench line of the Tihama type, went in across either zone between 11 and 22 September. Low: that smaller fighting positions are absent. The radar cannot rule them out. Unassessed: the stronger coastal change could hide some military activity among the farming. The imagery can't separate the two.

**Why it matters.** A negative finding was made usable: what was ruled out, at what size, against what control window (a late-August period with no front movement), and what remains unknown.

EXCERPT 8 **Six words, one root cause** `session 7e6be4f9 · 23 Sep 18:28Z · prompt 45`

ANALYST

`i only see a black screen`

CLAUDE

You opened the template (`earthwork_scan/wayback_manual_tracks_review.tpl.html`) instead of the built page. The template has no data in it, so it renders empty. […] That was my fault: I left the template as a `.html` file and it opened in the browser pane while I was writing it. To prevent a repeat: I renamed the template to `wayback_review_template.txt`, so it no longer looks like a page. If a copy of the template is opened anyway, it now shows a message pointing to `review.html` instead of a blank page.

**Why it matters.** Claude took responsibility for a confusing file layout instead of asking the analyst to try again, and fixed the structural cause.

EXCERPT 9 **"Actually download"** session 7e6be4f9 · 23 Sep 22:20–22:31Z · prompts 48, 49

ANALYST

I want the download to actually download, not to open the full-res image in a new tab. remove the high-res button, it's duplicated by the download button

CLAUDE

The cause: when the page is opened straight from disk, Chrome ignores the `download` attribute on local links. Because a JPG is something Chrome can display, it then just opens the image, which is what you saw. The fix: each tile also gets a GeoTIFF copy in a `downloads/` folder, and the buttons point at those. Chrome can't display TIFFs, so it always saves them.

ANALYST (SEVEN MINUTES LATER)

@"/Volumes/T7/yemen-maps/wayback_manual_tracks/review.html"
i want you to build in all of the images into this tool

CLAUDE

`review.html` now has every image built in: all 670 tiles at full resolution, plus their thumbnails. […] I ran the same GeoTIFF code outside the browser and read the result back with GDAL/rasterio: pixels and georeferencing matched the reference file exactly. Not checked: clicking Download with the page opened from disk in your own Chrome, since my browser can't open files from disk.

**Why it matters.** Two rounds of UI iteration moved from a workaround to a single 420 MB file that runs offline and builds GeoTIFFs in the browser. Note the final sentence: even at the end, the one thing Claude could not test is named.

EXCERPT 10 **Resolving a vague reference, and saying how it was read** `session a4e8eb74 · 23 Sep 01:59–02:23Z · prompt 39`

ANALYST

```
I want to you to rerun the latest iteration of the analysis on the original
region I requested
```

CLAUDE

*[first actions: read all memory files, list recent files and region configs, extract user prompts from past sessions, inspect the front2026 scripts.]*

I read "latest iteration" as the September front-construction method: Sentinel-1 and Sentinel-2 change since the offensive, compared against a late-August control, then dated pass by pass. If you meant rerunning the earlier trench-detection package on this area instead, say so and I'll run that.

**Why it matters.** A twenty-word prompt with two ambiguous references was resolved from project state in seconds, and the interpretation was stated so it could be corrected cheaply. The analyst's next prompt asked for a KML of the results.

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## 7. What went wrong

A methodology worth sharing includes the failures. These are the ones visible in the logs; none was fatal, and most were fixed within a turn or two. The last column is the takeaway to carry over.

| WHAT HAPPENED | SESSION | RESOLUTION | TAKEAWAY |
|---|---|---|---|
| OSM query returned HTTP 406 | d86eefd2 | Added a User-Agent header. | Public APIs need identification. |
| Named a World Bank dataset that did not exist | d86eefd2 | Retracted after searching (Excerpt 2). | Ask for sources, not recollection. |
| Mislabeled the trench as a wadi channel | d86eefd2 | Corrected next turn (Excerpt 1). | Allow revision; list corrections. |
| A script named `select.py` shadowed a Python standard-library module | d86eefd2 | Renamed. | Avoid stdlib names for scripts. |

| WHAT HAPPENED | SESSION | RESOLUTION | TAKEAWAY |
|---|---|---|---|
| Optical-first scan caught 1.3 km of a 9 km complex | d86eefd2 | Radar-first redesign. | Calibrate on a known positive early. |
| Hand trace of Site B was 50–100 m off the trench | 448c7361 | Re-snapped to 0.3 m imagery. | Validate labels before scoring against them. |
| Homebrew Python has no geo stack; only system Python 3.9 does. Earth Engine CLI crashes on 3.9 | 448c7361, 3981a117 | Recorded in memory; use the library, not the CLI. | Write environment quirks into memory on first contact. |
| Radar downloads at 0.1–0.4 MB/s | 448c7361, 264fa09b | Moved to Earth Engine. | Time a small read before committing. |
| Published a report as a claude.ai Artifact; analyst wanted offline HTML | 448c7361 | Saved as a feedback memory; later reports were offline. | State the delivery format up front. |
| Older Wayback releases lack zoom-18 tiles; mosaics came out black | 8d05f04e | Fell back to zoom 17 and tagged those frames. | Open the overview image before comparing. |
| Fusing four maps before extraction lost weak lines; learned ranker (24 positives) no better than hand score | 264fa09b | Per-map union; automatic fallback (Excerpt 6). | Let the evaluation veto the sophistication. |
| Chip IDs from two halves of the area collided and overwrote each other | a4e8eb74 | Prefixed IDs and rebuilt. | Split runs need distinct namespaces. |
| Browser pane cannot open `file://` paths or read the external drive | seven sessions | Mirror the page into the scratchpad and serve it over localhost. | Put the workaround in a note; it cost a few minutes each time. |
| Progress monitor emitted 88 wake-ups | 8d05f04e | Later monitors reported only start, end or error. | Monitor completion, not progress. |
| Template opened as if it were the page: "black screen" | 7e6be4f9 | Renamed template (Excerpt 8). | Do not leave build inputs looking like outputs. |
| Chrome ignores `download` on local files it can display | 7e6be4f9 | In-browser GeoTIFF (Excerpt 9). | Test the last step in the real browser. |
| Automatic screen ranked Marib's main 4.2 km ditch 454th of 1,138 | 044a49f0 | Found by eye on a wider-view pass; stated in the report. | Screening needs a person scanning the imagery behind it. |

> **THE RECURRING SHAPE**
>
> Almost every failure was caught by **looking at an output** (an image, a table, a rank) rather than by an error message. Claude made 195 Read calls, most on images it had just rendered, and kept finding things: a color bar over a panel, black mosaics, overlapping chart labels. The most expensive mistakes were the ones that produced a plausible number without anyone looking, which is why "look at the result" belongs in the prompt.

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## 8. Getting started: a playbook

### 8.1 Set up once

- **A project folder on a fast disk with room.** This one reached 32 GB. Claude moved the first 8.4 GB to an external SSD on day two, verified the copy by checksum and asked before deleting the original.
- **A working Python with the geospatial stack** (numpy, OpenCV, scikit-image, rasterio, scipy, matplotlib, scikit-learn). Tell Claude which interpreter; here the system Python 3.9 was the only one.
- **An Earth Engine account and cloud project.** Authenticate through the Python library, not the command line. Claude stored the project so scripts needed no arguments. Do not paste the token file or project credentials into chat; this paper's prompt ledger redacts the project ID.
- **Memory on.** The first notes written were the environment, a "state of the project" note and the deliverable format.
- **Optional skills** for house style: a paper template and a charting guideline.
- **RAM discipline.** With 8 GB, two full-frame rasters plus a download already paged; Claude tiled work and ran heavy jobs one at a time.

### 8.2 Prompt templates lifted from the ledger

Each is a real prompt with the specifics replaced by brackets. The last column is where to find the original in Appendix B.

| # | PATTERN | PROMPT (FILL THE BRACKETS) | LEDGER |
|---|---|---|---|
| 1 | **Anchor** | `can you use [overpass turbo or query OSM directly] to get data layers around [feature] in a [50]km radius from these coordinates? [lat, lon]` | #1 |

| # | PATTERN | PROMPT (FILL THE BRACKETS) | LEDGER |
|---|---|---|---|
| 2 | **Feasibility** | `is there a way to detect [the building of X] with [sensor] imagery? it begins at the precise point i've indicated` | #8 |
| 3 | **Scan** | `scan the rest of the [50km circle] for other new [earthworks], make a map for me superimposed on the latest [sentinel 2] image showing me candidate locations` | #9 |
| 4 | **Teach by example** | `there are major [X] here — can you adjust your detection process to prioritize things with a similar signature? [lat, lon]` | #10 |
| 5 | **Interview** | `I want you to make an analysis of [the imagery you've reviewed so far] from the perspective of a [role]. ask me questions to shape your analysis` | #21 |
| 6 | **Labels + package** | `[attach KMZ] I've hand labeled and identified more [X] in this region. Can you review this file and improve the signatures you use to detect these [X]? You may need multiple signature types. Use your best judgment - I want to apply lessons learned here to new regions. I want you to improve the signatures, improve detection, and then package the process so I can easily implement it in other regions. Ask me questions to help design the process` | #26 |
| 7 | **Look for change** | `can you look at the new areas of [fighting] and look for signs of new construction? show me on [sentinel 2] what you find` | #37 |
| 8 | **Reuse** | `I want you to rerun the latest iteration of the analysis on the original region I requested` | #39 |
| 9 | **Transfer** | `are there similar structures in the [marib frontline]?` | #40 |
| 10 | **Review file** | `create a [kml] for me to review / make an offline .html file for me to review` | #20, 41, 19 |
| 11 | **Consolidate** | `can you copy all of the [images] you've pulled for the manually defined tracks into a folder for me? I also want a CSV that shows the tile and the coordinates, and a label for the trench complex.` | #42 |

## 8.3 A CLAUDE.md to start with

Nothing like this existed in the project folder, and several lessons above would have been cheaper with it. A starting point derived from the record:

```
# Project rules (imagery analysis)
- Deliverables: self-contained offline HTML, KML/KMZ for Google Earth, GeoTIFF for QGIS.
  Do not publish to claude.ai unless asked. Keep a rebuild script beside every report.
- Dates: state the imagery CAPTURE date, never the release date.
- End every substantial result with: what you did NOT verify, and what would change your
conclusion.
- Give confidence levels with their basis. Do not attribute who built something beyond what
the evidence supports.
- Before any large download, time a small read and tell me the estimate.
- Wait on jobs with a monitor that fires on completion or error, not on progress.
- Ask before deleting, publishing or sending anything. Never put credentials in chat, logs or
files.
- Environment: use /usr/bin/python3 (has the geo stack); Earth Engine via the library, not the
CLI.
- Browser pane cannot open file:// or the external drive: copy the page to the scratchpad and
serve locally.
```

## 8.4 Running `trenchkit` on a new region

```
cp regions/template.json regions/my_region.json      # edit the bounding box, epochs and
Wayback settings
PYTHONPATH=. /usr/bin/python3 -m trenchkit.run regions/my_region.json --labels my_labels.kmz
```

Stages, in order: `grid, labels, fetch (S2/S1 or Earth Engine), detect, fuse, features, rank, review, wayback, evaluate`. Reported baselines to beat on the Tihama region: the top 25 candidates contain 15 % of label kilometers, the top 100 contain 19 %, and the top 400 contain 27 %. Every labeled complex reaches the top of the list, but the faintest trenches lie outside any candidate. Screening, not mapping, is what it is for.

## 8.5 A first-week plan

| DAY | DO | SUCCESS LOOKS LIKE |
|---|---|---|
| 1 | Use templates 1–3 on a place you know. End with "create a KML for me to review". | You open Claude's output in your own tool and find one error in it. |
| 2 | Hand-label 10–20 examples. Give the file to Claude with template 6. | Numbers that say which signals work, with the labels checked against imagery. |
| 3 | Ask for the review page with a verdict box per candidate. Review 50. | A verdict JSON you can feed back as new labels. |
| 4–5 | Point it at a second region. Write the notes and the CLAUDE.md. | A new session resumes cold and continues correctly. |

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## 9. Limits, ethics and confidence

### HONEST LIMITS OF THIS RECORD

- **Not verified by the author.** This paper was drafted by Claude from the session logs. It did not re-run the pipeline or re-view the imagery; findings are as reported in the sessions.
- **Files not opened in their target apps.** Claude reported that it validated the KML/KMZ files as XML but had not opened them in Google Earth, and that in-page downloads were tested against a local server but not from `file://` in the analyst's own Chrome.
- **Partial checks.** The Marib north bank was examined at 0.6 m along only 1.5 km of its ~11 km; the Al-Mashjah rim feature was dropped as unverified; 74 of 113 candidates in the Tihama rerun were flagged "not individually reviewed".
- **Imagery is older than the events.** The newest Wayback frames over these sites are from 2025 (Marib: February 2025). Sentinel-2 "after" epochs straddle the 10–11 September offensive, so anything dug in September is hidden in the medians.
- **Sources.** Context on the front came largely from outlets aligned with one side citing unnamed local sources. Claude flagged this and treated those items as leads.
- **Attribution** rested on geography and timing, not observed units, and was rated moderate at best.

### Dual use and responsibility

The same techniques that locate fortifications can be used by any party, and the coordinates in the underlying reports are precise. The analyst chose to keep exact locations and attributions in this record. Readers adapting the method should decide separately what they can responsibly publish, should not treat screening output as proof, and should preserve the rejected-example sections that make a result reviewable. No credentials, tokens or project identifiers are included; the ledger redactions are listed in Appendix D.

### What these numbers do and do not say about cost

Token counts are given raw: 1,056,709 output tokens and about 176 million cache-read tokens across all sessions. [3] The cache-read figure counts the same conversation context being re-read on every step, so it measures how long the sessions ran rather than how much text was processed. This paper does not convert usage to a price.

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## 10. Conclusion

In 50 hours, 41 short prompts and 989 tool calls took a single coordinate to dated trench complexes on the Tihama coast and at Marib, a reusable screening package, a fast way to review candidates in the analyst's own tools, and three applications of the method beyond the starting area. The gain came from the loop, not from any single clever prompt: frame, let Claude explore and build, make it check itself, receive a file you can open, review it, and store what was learned.

**First qualification.** The loop only converges when the human contributes ground truth. The 185 km of hand labels were the single most valuable input; without them Claude would have optimized against its own guesses. **Second qualification.** The record shows what Claude reported, and Claude reported on itself. The safeguards that made that credible, such as explicit unverified lists, rejected examples, confidence tiers and human approval of irreversible steps, were partly Claude's habit and partly the analyst's choices, and the safest course is to write them down as rules.

> THE STORY THAT DOESN'T FIT IN A TABLE
>
> The first request was for irrigation canals. The analyst wanted a map layer for Google Earth, and Claude produced 849 OSM features and a KML. The second request was for a World Bank dataset that did not exist, and Claude said so. By the tenth prompt the analyst was pointing at a coordinate and saying, in effect, **there are major trenches here, so learn what they look like**. Everything after that followed from a habit that started in the first hour: every answer was a file the analyst could open, question, and correct.
>
> *Short prompts, reviewable files, written-down rules. (◕‿◕✿)*

## Appendix A. Session table

Times are UTC as logged. "Typed" counts typed prompts only. Output tokens are summed from the assistant messages in each log.

| SESSION | TITLE (AS LOGGED) | START | END | TYPED | TOOL CALLS | OUTPUT TOKENS | MODEL |
|---|---|---|---|---|---|---|---|
| d86eefd2 | OSM canal/irrigation data query | 09-21 20:44 | 09-21 22:53 | 10 | 116 | 121,810 | Fable 5.1 |
| 007112a8 | Project SSD migration | 09-22 05:17 | 09-22 05:21 | 1 | 5 | 1,833 | Fable 5.1 |
| 75b23107 | Latest ERSI wayback image for trench detections | 09-22 05:24 | 09-22 07:04 | 4 | 42 | 32,136 | Fable 5.1 |
| 448c7361 | Military trenches detection signature | 09-22 05:53 | 09-22 07:47 | 6 | 132 | 202,534 | Fable 5.1 |
| c66378c1 | Military geospatial analysis: Sentinel/Wayback imagery | 09-22 07:08 | 09-22 07:49 | 1 | 103 | 148,442 | Fable 5.1 |
| 8d05f04e | Yemen trench complexes wayback imagery | 09-22 17:05 | 09-22 18:55 | 1 | 44 | 34,787 | Fable 5.1 |
| 264fa09b | Yemen trench detection signatures | 09-22 18:43 | 09-22 20:25 | 1 | 121 | 179,877 | Fable 5.1 |
| 7c12864d | Google Earth Engine API key access | 09-22 22:19 | 09-22 22:21 | 0 | 2 | 5,677 | Fable 5.1 |
| 3981a117 | Google Earth Engine API key access (fork) | 09-22 22:19 | 09-23 01:38 | 6 | 118 | 101,050 | Fable 5.1, Opus 5.5 |
| a4e8eb74 | Rerun analysis on original region | 09-23 01:59 | 09-23 02:27 | 2 | 50 | 36,413 | Opus 5.5 |
| 044a49f0 | Marib frontline structures | 09-23 02:18 | 09-23 04:24 | 1 | 148 | 80,946 | Opus 5.5 |
| 7e6be4f9 | Yemen trench wayback images export | 09-23 18:10 | 09-23 22:31 | 8 | 108 | 111,204 | Opus 5.5 |
| **Total** | | | | **41** | **989** | **1,056,709** | |

## Appendix B. Full prompt ledger

Every human input in the 12 sessions, in time order, verbatim. Four kinds appear: typed into the chat; terminal commands the analyst ran themselves through the app's shell shortcut; paste where the analyst typed a sentence and pasted an error; slash app commands; and form answer where the analyst answered a structured question form. Harness-generated messages (task notifications, skill text, "continue from where you left off") are excluded. The same ledger is in `PROMPTS.md` in the repository.

| # | TIME (UTC) | SESSION | KIND | VERBATIM TEXT |
|---|---|---|---|---|
| 1 | 09-21 20:44Z | d86eefd2 | typed | can you use overpass turbo or query OSM directly to get data layers around canal or irrigiation in a 50km radius from these coordiantes? 14.106096485172495, 43.099555754076974 |
| 2 | 09-21 20:45Z | d86eefd2 | typed | can you create a .kml file for me to open this in google earth? |
| 3 | 09-21 20:45Z | d86eefd2 | typed | can you pull the world bank tihama irrigation datasets for me too |
| 4 | 09-21 20:48Z | d86eefd2 | typed | is there new of houthis constructing military trenches in this area in local reporting? |
| 5 | 09-21 20:50Z | d86eefd2 | typed | Can you pull sentinel-2 imagery for this area and develop a timelapse for me between 2021 to present? only pictures without cloud cover 3 FPS |
| 6 | 09-21 21:16Z | d86eefd2 | typed | can you center the timelapse here? this point should be on the coast 14.10770228114205, 43.09836884866308 |
| 7 | 09-21 21:21Z | d86eefd2 | typed | this is great. using the same long and lat, can you do an NDVI analysis from 2020 - present to show impact on agriculture? make an assessment for me based on this analysis |
| 8 | 09-21 21:32Z | d86eefd2 | typed | is there a way to detect the building of the military trench with sentinel imagery? it begins at the precise point i've indicated |
| 9 | 09-21 21:50Z | d86eefd2 | typed | scan the rest of the 50km circle for other new earthworks, make a map for me superimposed on the latest sentinel 2 image showing me candidate locations |

| # | TIME (UTC) | SESSION | KIND | VERBATIM TEXT |
|---|---|---|---|---|
| 10 | 09-21 22:16Z | d86eefd2 | typed | there are major trenches and earthworks here - can you adjust your detection process to proritize things with a similar signature? 14.056300910076873, 43.431572797625435 |
| 11 | 09-22 05:17Z | 007112a8 | typed | I want you to move the project to the attached SSD |
| 12 | 09-22 05:24Z | 75b23107 | typed | for the latest round of trench detections, I want you to grab the latest ERSI wayback image available of that area for me to review |
| 13 | 09-22 05:51Z | 75b23107 | typed | i'd like for you to make the traces more easily seen on the contact sheet .jpg. increase the resolution as well to make it easier for me to review |
| 14 | 09-22 05:52Z | 75b23107 | typed | pull an older wayback release for A so I can compare before/after |
| 15 | 09-22 05:53Z | 448c7361 | typed | I want you to see if you can improve the detection signature for these military trenches, berms and other construction we are trying to detect via sentinel 1 and sentinel 2 data. |
| 16 | 09-22 05:56Z | 448c7361 | typed | B is the clearest example of known constructed trenches I am interested in. they tend to be wider |
| 17 | 09-22 06:47Z | 75b23107 | typed | @"/Users/brian/Downloads/Yemen Trenches.kmz" pull the imagery with before and after for the points specified in this kml - |
| 18 | 09-22 06:55Z | 448c7361 | typed | keep going |
| 19 | 09-22 06:59Z | 448c7361 | typed | pull wayback imagery for the two new B-type candidates, then make an offline .html file for me to review |
| 20 | 09-22 07:06Z | 448c7361 | typed | create a kml for me to review the latest round as well |
| 21 | 09-22 07:08Z | c66378c1 | typed | I want you to make an analysis of the sentinel 1, 2 and wayback imagery you've reviewed so far from the perspective of a military geospatial analyst. ask me questions to shape your analysis |

| # | TIME (UTC) | SESSION | KIND | VERBATIM TEXT |
|---|---|---|---|---|
| 22 | 09-22 07:13Z | c66378c1 | form answer | QUESTIONS: What kind of analytical product should this be? \| How far should I go on attribution and context beyond what the imagery shows? \| Which features should the analysis cover? \| What format do you want the analysis delivered in?<br>ANSWERS: "What kind of analytical product should this be?"="All three in one document", "How far should I go on attribution and context beyond what the imagery shows?"="Attribute and assess intent", "Which features should the analysis cover?"="Site B and B-type look-alikes,Site A trench complex,Ruled-out set and civil change", "What format do you want the analysis delivered in?"="HTML analysis paper (Recommended)". |
| 23 | 09-22 07:16Z | c66378c1 | form answer | QUESTIONS: Do you hold ground truth on who built or holds sites A and B? This anchors the attribution section. \| Who is the reader of the paper? \| Should I spend session time on extra collection before writing?<br>ANSWERS: "Do you hold ground truth on who built or holds sites A and B? This anchors the attribution section."="No, assess from open sources", "Who is the reader of the paper?"="Analyst peers or OSINT researchers (Recommended)", "Should I spend session time on extra collection before writing?"="write from what is on disk now, defer the other options to be done later after I review what's on disk". |
| 24 | 09-22 07:25Z | 448c7361 | typed | for the off line review i want the full information from this latest round |
| 25 | 09-22 17:05Z | 8d05f04e | typed | @"/Users/brian/Downloads/Yemen identified trenches.kmz"<br>I've manually mapped out the trench complexes. Can you obtain the wayback imagery with a before/after for these areas marked? |

| # | TIME (UTC) | SESSION | KIND | VERBATIM TEXT |
|---|---|---|---|---|
| 26 | 09-22 18:43Z | 264fa09b | typed | @"/Users/brian/Downloads/Yemen identified trenches.kmz"<br>I've hand labeled and identified more trench systems in this region. Can you review this KML file and improve the signatures you use to detect these earthworks? You may need multiple signature types. Use your best judgment - I want to apply lessons learned here to new regions in Yemen. I want you to improve the signatures, improve detection, and then package the process so I can easily implement it in other regions. Ask me questions to help design the process |
| 27 | 09-22 22:19Z | 3981a117 | typed | how can i get to my google earth engine api key |
| 28 | 09-22 22:20Z | 3981a117 | typed | I just registered |
| 29 | 09-22 22:21Z | 3981a117 | terminal | /usr/bin/python3 -m pip install --user earthengine-api |
| 30 | 09-22 22:21Z | 3981a117 | terminal | /usr/bin/python3 -m ee.cli authenticate |
| 31 | 09-22 22:21Z | 3981a117 | paste | I got this error when I tried step 2 [pasted terminal error output omitted] |
| 32 | 09-22 22:22Z | 3981a117 | terminal | /Users/brian/Library/Python/3.9/bin/earthengine authenticate |
| 33 | 09-22 22:22Z | 3981a117 | terminal | /usr/bin/python3 -W ignore -c "import ee; ee.Authenticate()" |
| 34 | 09-22 22:24Z | 3981a117 | typed | [GCP_PROJECT_ID] |
| 35 | 09-22 22:25Z | 3981a117 | typed | pull Sentinel-1 stacks over sites A and B |
| 36 | 09-22 22:41Z | 3981a117 | slash | /model claude-opus-5-5 |
| 37 | 09-23 00:16Z | 3981a117 | typed | can you look at the new areas of fighting and look for signs of new construction? show me on sentinel 2 what you find |
| 38 | 09-23 01:38Z | 3981a117 | typed | summarize for me your assessment based on the sentinel 1 data analysis |
| 39 | 09-23 01:59Z | a4e8eb74 | typed | I want to you to rerun the latest iteration of the analysis on the original region I requested |
| 40 | 09-23 02:18Z | 044a49f0 | typed | are there similar structures in the marib frontline? |

| # | TIME (UTC) | SESSION | KIND | VERBATIM TEXT |
|---|---|---|---|---|
| 41 | 09-23 02:26Z | a4e8eb74 | typed | can you create a kml for me to review |
| 42 | 09-23 18:10Z | 7e6be4f9 | typed | @"/Users/brian/Downloads/Yemen Trenches.kmz" @"/Users/brian/Downloads/Yemen identified trenches.kmz"<br>Can you copy all of the wayback images you've pulled for the manually defined tracks into a folder for me? I also want a CSV that shows the tile and the coordinates, and a label for the trench complex.<br><br>I also want you to be sure that the images associated with these two KML files are represented. We've already pulled the images so you shouldn't have to redownload it |
| 43 | 09-23 18:17Z | 7e6be4f9 | typed | can you create a well designed .html file for me to review what we have? I also want you to fill the gaps and download those tiles |
| 44 | 09-23 18:27Z | 7e6be4f9 | typed | can you build the html with the partial image coverage for now? |
| 45 | 09-23 18:28Z | 7e6be4f9 | typed | i only see a black screen |
| 46 | 09-23 22:15Z | 7e6be4f9 | typed | I want you to add a way to easily download the high res version in the .html |
| 47 | 09-23 22:17Z | 7e6be4f9 | typed | can you do the middle trenches too? |
| 48 | 09-23 22:20Z | 7e6be4f9 | typed | I want the download to actually download, not to open the full-res image in a new tab. remove the high-res button, it's duplicated by the download button |
| 49 | 09-23 22:27Z | 7e6be4f9 | typed | @"/Volumes/T7/yemen-maps/wayback_manual_tracks/review.html"<br>i want you to build in all of the images into this tool |

## Appendix C. Deliverables inventory

| FILE OR FOLDER | WHAT IT IS | MADE IN |
|---|---|---|
| `osm_irrigation/*.geojson`, `yemen_irrigation_osm.kml` | 849 OSM features in six layers; KML for Google Earth | d86eefd2 |
| `s2_timelapse/`, `ndvi_analysis/` | Sentinel-2 timelapses (173 and 144 frames); NDVI change analysis and figures | d86eefd2 |
| `earthwork_scan/` | Detection scripts (68 files, ~3,200 lines), candidate maps and KMLs, Wayback pulls, label evaluations | d86eefd2 → 264fa09b |
| `tihama_earthworks_analysis.html` | Military-analyst paper on sites A, B, look-alikes and civil change (rebuild kit in `paper_assets/`) | c66378c1 |
| `earthwork_scan/review_v3_full.html`, `review_btype_candidates.html` | Offline reviews of all 57 v3 candidates and two B-type look-alikes, with verdict boxes | 448c7361 |
| `wayback_before_after_identified_trenches.html` | Before / mid / after Wayback sheets for the hand-mapped complexes (29 MB, grown to include the middle complex) | 8d05f04e, 7e6be4f9 |
| `trenchkit/` | Region-agnostic pipeline: 16 modules (989 lines), README, region configs | 264fa09b |
| `trench_signature_review.html`, `work/…/fused_review.html` | What the labels showed; 100-candidate review page with epoch chips and Wayback frames | 264fa09b |
| `sep2026_front_construction.html`, `front2026/` | Kahbub tracks and the negative findings; kit for S1/S2 change with a control window | 3981a117 |
| `tihama_sep2026_rerun.html`, `tihama2026/`, `tihama_sep2026_review.kmz` | Rerun on the original area; 113-candidate review KMZ | a4e8eb74 |
| `marib_front_earthworks.html`, `marib_confirmed_lines.kml` | Raghwan ditch network and north bank; confirmed lines for Google Earth | 044a49f0 |
| `wayback_manual_tracks/` | 670 one-kilometer tiles, CSV with coordinates and complex labels, gap list, comparison sheets, and a 420 MB self-contained `review.html` | 7e6be4f9 |
| `memory (10 notes)` | Environment, project status, toolkit descriptions, deliverable preference, one per major result | across sessions |

## Appendix D. Method and measurement notes

- **Sources.** The session logs stored by Claude Code for two project folders: the original `~/Documents/yemen-maps` (2 sessions, 21–22 Sep) and its copy on the SSD (10 sessions, 22–23 Sep). The session that produced this paper, and one that began after 23 September, are excluded.
- **Counting rules.** A "typed prompt" is a user message with typed text that is not tool output, harness-generated, a slash command, a terminal shortcut, or a form answer. A "tool call" is an assistant tool-use block; sub-agent calls would be included but there were none. The two-turn fork stub (7c12864d) repeats the first prompt of 3981a117 and appears once in the ledger.
- **Tokens.** Summed once per assistant message ID from the usage fields: 1,056,709 output, 3,241,485 cache-creation, 175,924,494 cache-read. These are the logs' own figures.
- **Redactions.** One Google Cloud project ID replaced by `[GCP_PROJECT_ID]` (prompt 34). One OAuth authorization URL in a terminal output was not reproduced. Pasted terminal output in prompt 31 is omitted. No other prompt text was altered.
- **Quotes.** Claude's quotations are excerpts of its replies; "…" marks omissions, "[…]" in brackets marks summarized material, and kaomoji decorations were dropped. User text is unmodified.
- **Result figures** (recall, ranks, lengths, dates) are copied from Claude's summaries in the sessions cited, not recomputed. Their definitions are in the repository's `trenchkit/README.md` and the memory notes.
- **The offline-HTML preference.** The exact words "i want an offline html not an artifact" are recorded in the feedback note the project saved on 22 Sep; the original message is not among the retained logs, so it is not in the ledger.
- **Rebuild.** `prep_data.py` extracts the ledger and statistics from the logs; `build_case_study.py` assembles this page and, with `--pdf`, prints it with headless Chrome.

1. Timestamps are in UTC as stored in the logs. Because sessions sometimes overlapped and some waited on downloads, the times are best read as a sequence of events rather than as working hours.
2. Claude's replies in this project often ended with a kaomoji. They are omitted from quotes so that citations stay clean.
3. Usage is reported as logged per assistant message. Cache-read tokens are re-reads of prior context at each step; they are not an amount of new text and should not be added to output tokens as a measure of work.